

GII Index Reduction Analysis Report

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1 Executive Summary

This report documents a method for determining how far the number of indicators in the Global Innovation Index (GII) 2025 can be reduced without significantly compromising the index’s discriminatory power. The method was previously applied to the EU JRC’s Sustainable and Inclusive Well-being Index (Benczur et al. 2024; Boskovic et al. 2024); this report replicates it on GIJ’s 78 Level-1 indicators observed for 139 countries/economies.

Two perspectives are evaluated: a **country perspective** (do countries end up the same distance apart using a subset as with the full indicator set?) and a **variable perspective** (does a subset preserve the PCA structure of the full set?). The two perspective rely on complementary metrics. The country perspective is based on a Spearman rank correlation between full- and subset-distance matrices; the variable perspective is based on r_m^2 - the percentage of the full PCA’s variance structure recovered by the subset.

Both perspectives require the analyst to choose a subset of indicators. Subsets of 21 and 42 indicators are considered. As the GIJ has 21 sub-pillars, 21 is the minimum needed to cover every concept the index aims to measure.

Random selection provides the baseline: the best subset drawn from 1000 random selections at a given size, ignoring the conceptual structure entirely. Against it, four structured allocation rules are evaluated. Flat allocation assigns the same number of indicators to every group, either 3 per pillar or 1 per sub-pillar, both yielding 21 indicators. Proportional allocation instead sets each group’s budget in proportion to its share of the index, on the premise that a group’s indicator count reflects its importance or the difficulty of measuring it; this is applied across pillars and across sub-pillars at 42 indicators, and across pillars again at 21 so that the

two rules can be compared at a common budget. Proportional allocation across sub-pillars at 21 indicators is not evaluated separately, because that apportionment reduces to exactly one indicator per sub-pillar and is therefore identical to the corresponding flat rule.

The headline recommendation is that the “one indicator per sub-pillar” set of **21- indicators comes close to reproducing the PCA structure** ($r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full PCA’s variance structure) while also closely reproducing the distance between countries (Rand Index 0.886 against the full-data country clustering). A larger 42-indicator subset, built by allocating indicators proportionally to each pillar’s actual size rather than a flat count per pillar, reaches $r_m^2 = 92.6\%$ and a rank correlation of 0.955 with the full country-distance matrix. Both are documented below alongside the flat and proportional strategies they were chosen from.

Relative to random subsets of matched size, exhaustive within-group search yields a modest advantage when indicators are allocated at pillar level and none when they are allocated at sub-pillar level. This pattern holds at both the 21- and 42-indicator budgets, and under both metrics.

2 Introduction

When creating indicator dashboards and composite indices, indicator selection can be daunting. Stakeholders, statistics and conceptual relevance often diverge in recommendations or build up a large set of indicators that is impractical. A key statistical question is: how much of the information content (“variation”) of the full indicator set is preserved in a candidate subset?

This analysis applies two methodological approaches to the GII 2025 data (78 Level-1 indicators, 139 countries):

1. **Country Perspective:** Assessing how similar countries are using all indicators versus a subset, using distance matrices and clustering techniques
2. **Variable Perspective:** Using Principal Component Analysis (PCA) to measure how much information a subset captures compared to the full set

2.1 Finding a suitable reduced set

Composite indices such as the GII combine two different measurement logics. At the upper levels - pillars, sub-indices, the index itself - the structure is *formative*: each pillar captures a distinct dimension of innovation performance, and the index’s meaning depends on all of them being represented; dropping a pillar entirely would change what the index measures, not just how precisely it measures it. Within a sub-pillar, by contrast, the indicators are closer to *reflective* measures of a single underlying concept - largely interchangeable manifestations of

the same thing, where one or two well-chosen indicators can stand in for the group without changing what is being measured.

This distinction motivates the reduction strategy used throughout this analysis: every pillar and sub-pillar is retained in every candidate subset, preserving the formative structure and therefore the index’s meaning, while indicator reduction happens only *within* each group, by selecting which reflective indicator(s) best represent it. We apply two strategies to decide *how many* reflective indicators to draw from each formative group. The first is to select one indicator per sub-pillar and the second is to select a number per pillar proportional to it’s original size. The metrics for these two strategies ignore correlations between pillars or sub-pillars. However, these cannot be incorporated without also expanding the search to be exhaustive.

We have not implemented an exhaustive search for the best subset. Even restricted to searching *within* a single pillar, exhaustive search grows quickly. Selecting the best possible subset of size k out of p indicators requires evaluating $C(p, k) = \frac{p!}{k!(p-k)!}$ combinations. The Institutions pillar (IN.1) has 6 indicators, so choosing 2 of them already means 15 possible combinations. Multiplied across all 7 pillars, the number of combinations for a subset is prohibitive. This analysis searches exhaustively within each pillar or sub-pillar rather than across the whole index at once. The table below shows how many combinations remain even after allocating a 20-indicator budget proportionally across pillars.

Table 1: Unique combinations per pillar of GII when allocating a 20-indicator budget proportionally

Pillar	Indicators in Pillar	Allocated (budget=20)	Combina- tions
Institutions	6	1	6
Human capital and research	12	3	220
Infrastructure	9	2	36
Market sophistication	11	3	165
Business sophistication	15	4	1365
Knowledge and technology outputs	14	4	1001
Creative outputs	11	3	165

Even after this proportional allocation collapses the problem to one pillar at a time, there are still 1,767,710,124,180,000 combinations to explore in total. Underscoring why within-group exhaustive search is the only tractable approach.

3 Methodology

3.1 Statistical Methods Explanation

The variable perspective uses a subset-selection approach based on Principal Component Analysis (PCA). Rather than simply ranking individual variables by importance, this method evaluates entire subsets of variables according to how well they reproduce the PCA structure of the full dataset.

The criterion, r_m , was introduced by McCabe (1984) as one of several optimality criteria for selecting “principal variables”. In contrast to principle components, principle variables are a subsets of the original variables that best capture variation in the original set. The method was developed in order to select indicators which could function as a principle component while remaining interpretable. Cadima and Jolliffe (2001) formalize this same criterion as one of two indicators for how well a subset of variables approximates a full PCA, and show that r_m^2 is exactly the percentage of total variance in the original data accounted for by regressing the full data on the chosen subset. This is the interpretation of r_m^2 used throughout this report. Both papers show, that choosing variables by their PC loadings alone frequently produces subsets that are suboptimal compared to subsets chosen by directly maximizing r_m .

A key advantage of this method is its built-in protection against redundancy. For example, if multiple indicators are highly correlated, including one in the subset provides most of the information, and adding the others contributes relatively little additional ability to reconstruct the principal components.

The mathematical formula for the selection criterion is:

$$r_m = \sqrt{\frac{\sum_{i=1}^p \lambda_i r_{mi}^2}{\sum_{j=1}^p \lambda_j}}$$

where: - λ_i is the eigenvalue of principal component i (PC_i) from the full dataset, representing the variance explained by that component. The subscript m indexes a candidate subset - r_{mi}^2 is the coefficient of determination (R^2) between principal component i and the candidate subset of variables. Since PC_i is a linear combination of the original variables, the R^2 of the subset measures how well the principal component can be reconstructed from the subset.

Conceptually, this formula computes a weighted average of how well the subset predicts each principal component, with weights given by the variance explained by each component, then normalizes by the total variance. The resulting r_m^2 value can be interpreted as the percentage of total variation explained by the subset, providing an intuitive measure of information retention.

This method assumes that “information” is defined by the variance structure captured by PCA, which is appropriate for statistical data reduction. However, it may downweight variables with

co-variance but high policy relevance, which is an important consideration for substantive index development (c.f.(Algursan et al. 2026)).

3.1.1 The Country Perspective's Distance-Based Criterion

The country perspective asks assess whether a subset preserves how similar countries are judged to be relative to one another. For each candidate subset, the pairwise Euclidean distance between every pair of countries is computed using only that subset's indicators, and the ranking of those pairwise distances is compared to the ranking obtained from the full indicator set, via Spearman rank correlation. A correlation near 1 means the subset reproduces almost the same country-similarity ordering as the full index.

3.1.2 Step-wise Local Search

Exhaustive search identifies the single best-scoring subset within a group, but it is only tractable because it is applied separately to each pillar or sub-pillar; the criterion it maximizes for a given group is therefore local to that group's own indicators, rather than the full index. A step-wise (greedy) local-search procedure offers an alternative that scales to a group or a budget too large to search exhaustively, and that optimizes a candidate subset's score against the FULL indicator set directly, rather than against any single group in isolation. It applies equally to either selection criterion - the country perspective's rank correlation or the variable perspective's r_m - and to any allocation of indicators to groups.

The procedure proceeds as follows:

1. **Initialization.** For every group, the number of indicators specified by that group's allocation is drawn at random, without replacement, from the indicators belonging to that group. Together these draws form the initial candidate subset.
2. **Scoring.** The candidate subset is scored against the full indicator set - by the rank correlation of distance matrices, or by r_m against the full-dataset PCA - not against any single group's own indicators.
3. **Local search.** Each indicator currently in the subset is considered for replacement, in turn, by every other indicator belonging to the same group. A replacement is accepted whenever it raises the subset's full-dataset score, and search continues from the improved subset; a replacement that would introduce an indicator already present elsewhere in the subset is never considered, since a duplicated indicator would be double-counted in the underlying distance or PCA calculation.
4. **Convergence.** Step 3 is repeated, as a full pass over every group and every indicator in it, until a complete pass produces no further improvement, or a fixed maximum number of passes is reached.

Because the search starts from a random draw and only ever accepts a strictly improving substitution, it converges to a local optimum with respect to one-indicator-at-a-time swaps, which need not be the global optimum an exhaustive search would find were one computationally feasible. The procedure is therefore run from several independent random starting points, and the best-scoring outcome across those runs is retained.

3.2 Proportional Allocation Methodology

Both perspectives select candidate subsets by exhaustive search *within* each pillar or sub-pillar, then assemble the per-group picks into a full subset - consistent with the formative/reflective distinction. Every pillar and sub-pillar is represented in every candidate subset, and only the reflective indicators within each group are subject to selection. This requires deciding how many indicators to take from each group. Two allocation rules are used throughout this report:

- **Flat allocation of 21 indicators:** every pillar contributes (3) indicators, or every sub-pillar contributes 1 indicators. A simple strategy to implement, however, it ignores that GII's pillars differ in the number of indicators, from 6 indicators (Institutions) to 15 (Business sophistication), and its sub-pillars range from 2 to 5 indicators. A flat allocation over-represents small groups and under-represents large ones relative to their actual share of the index. This may limit the subset's utility statistically and leave out conceptual nuances of the GII measurement framework.
- **Proportional allocation of 21 or 42 indicators.** Each group's budget is set proportional to its share of all indicators, using a **largest-remainder (Hamilton) apportionment**. Every group first receives n indicators rounded down to the lowest integer. Any indicators left over are handed out one at a time to the groups with the largest fractional remainder. Unlike flat allocation, a group with a very small share of the index can receive **zero** indicators under this rule. The total budget used is **42** indicators (double the flat allocations' 21). Proportional allocation is applied across pillars and across sub-pillars at this budget, and across pillars again at **21** indicators, matching the flat allocations so that the effect of the allocation rule can be separated from the effect of the budget. No matching sub-pillar case exists: apportioning 21 indicators across 21 sub-pillars returns exactly one indicator each, which is the flat rule. This is also why the larger budget is set above the number of sub-pillars.

Each perspective searches on its own criterion, applied *locally* within a group. The variable perspective maximizes r_m against that group's own sub-PCA rather than the full 78-indicator PCA. The country perspective maximizes the Spearman rank correlation between that group's own distance matrix and the candidate subset's. The allocation rule and the selection criterion are independent choices, so all four allocation rules above are searched under both criteria. This local search is a tractable stand-in for an otherwise infeasible full search. The assembled subset is then re-scored against the full dataset to obtain the metrics reported in this document.

3.3 Data Description

The Global Innovation Index (GII) 2025 dataset ((WIPO) 2024) comprises 78 Level-1 indicators observed for 139 countries/economies, organized in a 5-level hierarchy: 78 indicators -> 21 sub-pillars -> 7 pillars -> 2 sub-indices (Inputs, Outputs) -> 1 overall index. Indicators with negative polarity (e.g. pupil-teacher ratio, applied tariff rate) are reversed before standardization, so that higher standardized values consistently indicate better innovation performance across all indicators. Missing values are imputed via 5-nearest-neighbors using the VIM (Kowarik and Templ 2016) package prior to z-scoring; see the Appendix for imputation statistics.

4 Key Findings

4.1 Country Perspective

4.1.1 Distance Matrix Analysis of Random Subsets

We selected random subset sizes ranging from 10% to 90% of all 78 indicators, while also selecting specifically 42, matching the proportional strategies' budget below. Each k subset was drawn 1000 times and compared to the full-dataset Euclidean distance matrix via Spearman rank correlation:

Table 2: Rank correlation of random subsets vs. the full distance matrix, by subset size k

Subset Size (k)	Mean Correlation	Median Correlation	Min Correlation	Max Correlation
8	0.678	0.687	0.383	0.824
15	0.784	0.789	0.626	0.878
23	0.847	0.851	0.726	0.918
31	0.891	0.893	0.808	0.936
38	0.920	0.922	0.856	0.955
42	0.932	0.935	0.872	0.962
46	0.946	0.948	0.898	0.971
54	0.963	0.965	0.902	0.983
62	0.978	0.979	0.945	0.991
69	0.988	0.989	0.957	0.996

Correlation rises smoothly and near-monotonically with subset size, from a mean of 0.678 at $k = 8$ to 0.988 at $k = 69$. This random-subset curve is the baseline to the local exhaustive-search strategies.

4.1.2 Flat Allocation

Finding: Flat allocation of 3 indicators per pillar reaches a rank correlation of 0.886 with the full distance matrix; flat allocation of 1 indicator per sub-pillar reaches 0.836. The comparable random baseline ($k = 23$) averages 0.847, with a maximum of 0.918 across 1000 draws.

The two allocation levels therefore diverge. Pillar-level allocation exceeds the random average by 0.039, while sub-pillar-level allocation falls 0.011 below it. Neither reaches the best random subset of the same size, which is expected: the search is confined within groups, so it cannot exploit the cross-group combinations available to an unconstrained draw. The constraint is accepted deliberately, since it is what guarantees every pillar and sub-pillar remains represented and the index retains its formative meaning.

4.1.3 Clustering Results: SIMPROF + Rand Index

Against the full-data country clustering, flat allocation by pillar recovers a Rand Index of 0.873 and flat allocation by sub-pillar 0.886. Both subsets therefore reproduce the full clustering to within roughly 13% of country pairs, and the two allocation levels are indistinguishable on this measure despite their 0.05 gap in rank correlation.

Table 3: Rand Index: full-data clustering vs. flat-allocation clustering (by pillar and by sub-pillar)

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Flat allocation (pillar)	0.873	78	21	Flat allocation (pillar)
Full vs. Flat allocation (sub-pillar)	0.886	78	21	Flat allocation (sub-pillar)

4.1.4 Proportional Allocation

We proportionally allocate 42 indicators, using the largest-remainder method described in Methodology above.

Finding: At the 42-indicator budget, proportional allocation across pillars achieves a rank correlation of 0.955, against 0.907 across sub-pillars. The random baseline at this size averages 0.932.

The advantage of pillar-level over sub-pillar-level allocation seen at 21 indicators persists at 42, and in the same direction: pillar-level allocation again exceeds the random average while

sub-pillar-level allocation falls below it. Doubling the budget raises every figure, but does not alter the ordering between the two allocation levels.

Applying the same proportional rule across pillars at the flat 21-indicator budget returns 0.887, against 0.886 for flat allocation at that budget. Separating the two effects this way shows the gain to be a function of the budget rather than of the rule.

Table 4: Rand Index: full clustering vs. proportional-allocation subset clustering (by pillar and by sub-pillar)

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Allocation Rule
Full vs. Proportional allocation (pillar)	0.892	78	42	Proportional allocation (pillar)
Full vs. Proportional allocation (sub-pillar)	0.863	78	42	Proportional allocation (sub-pillar)

Finding: Against the full-data country clustering, the 42-indicator proportional subset recovers a Rand Index of 0.892 when allocated across pillars and 0.863 when allocated across sub-pillars - against 0.873 and 0.886 for the two 21-indicator flat subsets.

All four values fall within 0.029 of one another, even though the subsets differ in size by a factor of two and span 0.119 in rank correlation. Recovery of the country clustering is therefore insensitive to both subset size and allocation level over this range: once every pillar and sub-pillar is represented, additional indicators sharpen the distance structure without redrawing the cluster boundaries.

4.1.5 Step-wise Local Search

The step-wise local-search procedure (Methodology) was run under the rank-correlation criterion for every allocation rule (flat, proportional), grouping level (pillar, sub-pillar) and budget (21, 42) considered above, with 5 independent random restarts per combination and the best-scoring restart retained.

Table 5: Step-wise search: rank correlation with the full distance matrix, by allocation

Level	Budget	Allocation Rule	Rank Correlation
Pillar	21	Flat	0.941
Pillar	21	Proportional	0.939
Pillar	42	Flat	0.976
Pillar	42	Proportional	0.976

Level	Budget	Allocation Rule	Rank Correlation
Sub-pillar	21	Flat	0.926
Sub-pillar	42	Flat	0.974
Sub-pillar	42	Proportional	0.977

Finding: At the 21-indicator budget, step-wise search reaches a rank correlation of 0.941 under the flat pillar allocation and 0.939 under the proportional pillar allocation - both above the exhaustive-search results for the same allocations (0.886 and 0.887 respectively) and above the random baseline at matched size (0.847). Under the flat sub-pillar allocation, step-wise search reaches 0.926, again exceeding both the exhaustive result (0.836) and the random baseline - reversing the earlier finding that sub-pillar-level allocation offers no advantage over chance. The same pattern - step-wise above both its exhaustive-search and random-baseline counterparts - holds at the 42-indicator budget across every allocation tested (Table above).

This gap is a direct consequence of what each method optimizes. Exhaustive search maximizes each group’s *local* score, computed only against that group’s own indicators; the full-dataset score reported for it is a byproduct, not the search objective (see Limitations). Step-wise search instead scores every candidate against the full dataset at every step, so it can find combinations that a group-by-group local search cannot see - at the cost of no longer guaranteeing that the result is the best subset achievable for that allocation, only a local optimum with respect to one-indicator-at-a-time swaps from a random start.

The table below breaks down, pillar by pillar, where the three approaches agree and diverge, for the 3-indicators-per-pillar budget flat allocation uses: flat allocation’s own exhaustive result, proportional allocation at the same matched budget, and step-wise search run under the flat allocation (so the step-wise-vs-exhaustive columns hold the per-pillar budget fixed). Every selected indicator appears in exactly one column: “Common to All Three” for indicators every approach picked, “Shared by Two” (labeled with which pair) for indicators exactly two agreed on, and one of the three “-only” columns for indicators picked by exactly one. A parallel breakdown at sub-pillar level is not shown: at this budget, proportional allocation across sub-pillars is mathematically identical to flat allocation (Methodology), so the three-way comparison would collapse to flat vs. step-wise alone, and the sub-pillar-level scores already appear in the table above.

Table 6: Country perspective: indicator overlap between flat allocation, proportional allocation, and step-wise search, by pillar (21-indicator budget)

Group	Within-Group Score (Flat)	Common to All Three	Shared by Two	Flat Only	Proportional Only	Step-wise Only
Institutions	0.930	Entrepreneurship policies and culture	Government effectiveness (Prop.+Step-wise)	Operational stability for businesses; Rule of law		Regulatory quality
Human capital and research	0.775		Gross expenditure on R&D (Flat+Prop.); Tertiary enrolment (Flat+Prop.); Tertiary inbound mobility (Flat+Prop.)			QS university ranking; Graduates in science and engineering; Researchers
Infrastructure	0.734	ICT access	Electricity output (Flat+Prop.)	GDP/unit of energy use		Logistics performance; ISO 14001 environmental certificates
Market sophistication	0.728	Domestic credit to private sector; Joint venture deals	Domestic industry diversification (Flat+Prop.)			Late VC deals intensity
Business sophistication	0.788		Knowledge-intensive employment (Flat+Prop.); Research talent (Flat+Prop.); University-industry R&D collaboration (Flat+Prop.)		Intellectual property payments	High-tech imports; Emerging knowledge workforce; ICT services imports

Group	Within-Group Score (Flat)	Common to All Three	Shared by Two	Flat Only	Proportional Only	Step-wise Only
Knowledge and technology outputs	0.792	PCT patents by inventor origin	High-tech exports (Flat+Prop.); ISO 9001 quality certificates (Flat+Prop.)		Intellectual property receipts	Scientific and technical articles; ICT services exports
Creative outputs	0.806	Industrial designs by origin; Top-level domains (TLDs)	Global brand value, top 5,000 (Flat+Prop.)			Cultural and creative services exports

4.2 Variable Perspective

4.2.1 PCA Analysis

The table below shows the number of principal components needed to capture specified levels of variance in the full 78-indicator dataset:

Table 7: Components needed for target variance levels (full PCA)

Variance Level	Components Needed
80%	17
90%	26
95%	36

Finding: The first 17 of 78 components already capture 80% of total variance, confirming substantial redundancy in the full indicator set - a necessary condition for indicator reduction to be viable at all. Note this is a property of the full PCA (unconstrained linear combinations of all 78 indicators), not directly comparable to how many *raw indicators* a subset needs.

4.2.2 r_m by Group and by Strategy

The exhaustive-search sections below each report a *global* r_m - the assembled subset’s fit to the full-dataset PCA - built up from a *local* r_m computed separately for every pillar or sub-pillar. This section collects both views into two reference tables before the individual strategies are discussed one at a time.

The table below reports flat allocation’s local r_m - the score its exhaustive search actually maximizes - for every pillar and every sub-pillar (the same per-pillar values that also appear as the “Within-Group Score (Flat)” column of the indicator-overlap table under Step-wise Local Search, below):

Table 8: Within-group r_m (flat allocation’s exhaustive search), by pillar and sub-pillar

Level	Group	r_m
Pillar	Institutions	0.965
Pillar	Human capital and research	0.757
Pillar	Infrastructure	0.784
Pillar	Market sophistication	0.764
Pillar	Business sophistication	0.746
Pillar	Knowledge and technology outputs	0.702
Pillar	Creative outputs	0.697

Level	Group	r_m
Sub-pillar	Institutional environment	0.938
Sub-pillar	Regulatory environment	0.975
Sub-pillar	Business environment	0.834
Sub-pillar	Education	0.626
Sub-pillar	Tertiary education	0.615
Sub-pillar	Research and development (R&D)	0.793
Sub-pillar	Information and communication technologies (ICTs)	0.877
Sub-pillar	General infrastructure	0.647
Sub-pillar	Ecological sustainability	0.588
Sub-pillar	Credit	0.650
Sub-pillar	Investment	0.750
Sub-pillar	Trade, diversification and market scale	0.639
Sub-pillar	Knowledge workers	0.802
Sub-pillar	Innovation linkages	0.727
Sub-pillar	Knowledge absorption	0.508
Sub-pillar	Knowledge creation	0.638
Sub-pillar	Knowledge impact	0.592
Sub-pillar	Knowledge diffusion	0.649
Sub-pillar	Intangible assets	0.613
Sub-pillar	Creative goods and services	0.550
Sub-pillar	Online creativity	0.639

4.2.3 Comparison of flat and proportional strategies in global r_m

The table below instead combines the *global* r_m reported by each of the flat and proportional strategies discussed in the following sections into a single comparison (the step-wise alternative to these is introduced later, and appears in the Cross-Allocation Comparison table with the rest):

Table 9: Global r_m (assembled subset vs. full-dataset PCA), by strategy

Strategy	Indicators	r_m	r_m^2 (%)
Flat allocation (sub-pillar, headline)	21	0.826	68.2
Flat allocation (pillar)	21	0.854	72.9
Proportional allocation (pillar)	42	0.963	92.6
Proportional allocation (sub-pillar)	42	0.946	89.5

Finding: At 42 indicators, proportional allocation by pillar reaches $r_m = 0.963$ ($r_m^2 = 92.6\%$) and proportional allocation by sub-pillar reaches $r_m = 0.946$ ($r_m^2 = 89.5\%$) - both a large step

up from the 21-indicator flat allocations, as expected with double the budget.

4.2.4 Step-wise Local Search

The same step-wise local-search procedure (Methodology) was run under the r_m criterion, for every allocation rule, grouping level and budget considered above, with 5 independent random restarts per combination and the best-scoring restart retained.

Table 10: Step-wise search: r_m against the full-dataset PCA, by allocation

Level	Budget	Allocation Rule	r_m
Pillar	21	Flat	0.885
Pillar	21	Proportional	0.885
Pillar	42	Flat	0.968
Pillar	42	Proportional	0.971
Sub-pillar	21	Flat	0.878
Sub-pillar	42	Flat	0.966
Sub-pillar	42	Proportional	0.967

Finding: At 21 indicators, step-wise search reaches $r_m = 0.885$ under the flat pillar allocation and 0.885 under the proportional pillar allocation - both above the corresponding exhaustive-search results (0.854 and 0.86) and above the random-subset baseline at matched size (0.857). Under the flat sub-pillar allocation, step-wise search reaches $r_m = 0.878$, again exceeding both the exhaustive result (0.826) and the random baseline - reversing the earlier finding that, at this budget, the constrained search offers no advantage over chance on the PCA criterion. The same pattern holds at 42 indicators across every allocation tested (Table above), for the same reason given in the country perspective: step-wise search directly optimizes the full-dataset r_m at every step, where exhaustive search optimizes each group’s local r_m and is only re-scored against the full PCA afterwards (Limitations).

As in the country perspective, the table below compares flat allocation’s exhaustive result, proportional allocation at the matched budget, and step-wise search run under the flat allocation, pillar by pillar, with every selected indicator accounted for in exactly one of the “Common to All Three”, “Shared by Two”, or single-approach “-only” columns.

Table 11: Variable perspective: indicator overlap between flat allocation, proportional allocation, and step-wise search, by pillar (21-indicator budget)

Group	Within-Group Score (Flat)	Common to All Three	Shared by Two	Flat Only	Proportional Only	Step-wise Only
Institutions	0.965	Entrepreneurship policies and culture	Policy stability for doing business (Flat+Step-wise); Regulatory quality (Flat+Step-wise)		Government effectiveness	
Human capital and research	0.757	Expenditure on education	Gross expenditure on R&D (Flat+Prop.); Tertiary enrolment (Flat+Prop.)			QS university ranking; Researchers
Infrastucture	0.784	Low-carbon energy use	ICT access (Flat+Prop.); Gross capital formation (Flat+Step-wise)			Electricity output
Market sophistication	0.764	Joint venture deals	Domestic credit to private sector (Flat+Prop.); Domestic market scale (Flat+Prop.)			Late VC deals intensity; Market capitalization
Business sophistication	0.746		Females employed w/advanced degrees (Flat+Prop.); High-tech imports (Flat+Prop.); University-industry R&D collaboration (Flat+Prop.); FDI net inflows (Prop.+Step-wise)			Emerging knowledge workforce; Intellectual property payments

Group	Within- Group Score (Flat)	Common to All Three	Shared by Two	Flat Only	Pro- por- tional Only	Step-wise Only
Knowl- edge and technol- ogy outputs	0.702		Software spending (Flat+Prop.); Production and export complexity (Flat+Prop.); PCT patents by inventor origin (Prop.+Step-wise); Utility models by origin (Prop.+Step-wise)	Patents by ori- gin		ISO 9001 quality certificates
Creative outputs	0.697		Entertainment and media market (Flat+Prop.); National feature films (Flat+Prop.); Industrial designs by origin (Flat+Prop.)			Intangible asset intensity, top 15; Mobile app creation; Creative goods exports

4.2.5 Flat-Allocation Sweep: 1-4 Indicators per Pillar

To see how r_m grows as the flat per-pillar allocation increases, the flat allocation by pillar search was repeated for 1, 2, 3, and 4 indicators per pillar:

Table 12: r_m as the flat per-pillar allocation increases

Indicators per Pillar	r_m	Total Indicators
1	0.705	7
2	0.802	14
3	0.854	21
4	0.900	28

Finding: r_m rises from 0.705 at 7 variables (1/pillar) to 0.9 at 28 variables (4/pillar) - a steep gain per additional variable early on, consistent with the PCA finding above that a small number of dimensions already captures most of the variance structure.

4.2.6 Random-Subset r_m Curve vs. #PCs Needed

For the same subset sizes used in the country perspective's Fig 2, r_m was computed for 20 random subsets per size. r_m is far more expensive per evaluation than the rank-correlation metric. It requires regressing all 78 principal components on the candidate subset, so fewer random draws were used per size here than Table 2's 1,000.

Table 13: Random-subset r_m by size, vs. cumulative variance explained by the first k principal components

Subset Size (k)	r_m	Subset Variance Captured	PCA Variance (k components)
8	0.695	0.484	0.649
15	0.791	0.626	0.785
23	0.857	0.734	0.878
31	0.900	0.811	0.931
38	0.925	0.856	0.959
42	0.941	0.885	0.969
46	0.951	0.904	0.978
54	0.969	0.940	0.989
62	0.983	0.966	0.996
69	0.992	0.984	0.999

Finding: At $k = 21$ (interpolating between $k = 15$ and $k = 23$ above), a random subset averages roughly the same r_m as the exhaustive flat-allocation results report (0.854 by pillar /

0.826 by sub-pillar) - so at this budget the constrained search offers no advantage over chance on the PCA criterion, in contrast to the country perspective, where pillar-level allocation exceeded its random baseline. The PCA-variance column is consistently higher than a same-sized *indicator* subset's r_m^2 at every k. Since the first k principal components are unconstrained linear combinations of all 78 indicators, a subset is restricted to k raw indicators. The resulting gap in explained variance is the cost of requiring the final subset to be interpretable indicators rather than statistical composites.

4.2.7 Headline Recommended Subset

The 21-indicator subset from flat allocation by sub-pillar above, with full indicator names:

Table 14: Headline subset: one reflective indicator per sub-pillar

Sub-Pillar	Indicator	r_m of Best Subset
Institutional environment	Operational stability for businesses	0.938
Regulatory environment	Rule of law	0.975
Business environment	Entrepreneurship policies and culture	0.834
Education	School life expectancy	0.626
Tertiary education	Tertiary enrolment	0.615
Research and development (R&D)	Gross expenditure on R&D	0.793
Information and communication technologies (ICTs)	ICT access	0.877
General infrastructure	Logistics performance	0.647
Ecological sustainability	ISO 14001 environmental certificates	0.588
Credit	Finance for startups and scale-ups	0.650
Investment	Joint venture deals	0.750
Trade, diversification and market scale	Domestic industry diversification	0.639
Knowledge workers	Knowledge-intensive employment	0.802
Innovation linkages	University–industry R&D collaboration	0.727
Knowledge absorption	Research talent	0.508
Knowledge creation	PCT patents by inventor origin	0.638
Knowledge impact	High-tech manufacturing	0.592

Sub-Pillar	Indicator	r_m of Best Subset
Knowledge diffusion	Production and export complexity	0.649
Intangible assets	Industrial designs by origin	0.613
Creative goods and services	Entertainment and media market	0.550
Online creativity	GitHub commits	0.639

Finding: This subset captures r_m^2 68.2% of the full dataset’s PCA variance structure and recovers a Rand Index of 0.886 against the full-data country clustering, while cutting the indicator count from 78 to 21 (a 73% reduction).

5 Cross-Allocation Comparison

The tables below collapse every allocation rule, grouping level and budget considered above - under both exhaustive search and step-wise search - into a single comparison per perspective.

Table 15: Country perspective: allocation rules compared, by level, budget and search method

Level	Allocation Rule	Indicators	Rank Correlation	Search Method
Pillar	Flat	21	0.886	Exhaustive search
Pillar	Flat	21	0.941	Step-wise search
Pillar	Proportional	21	0.887	Exhaustive search
Pillar	Proportional	21	0.939	Step-wise search
Pillar	Flat	42	0.976	Step-wise search
Pillar	Proportional	42	0.955	Exhaustive search
Pillar	Proportional	42	0.976	Step-wise search
Sub-pillar	Flat	21	0.836	Exhaustive search
Sub-pillar	Flat	21	0.926	Step-wise search
Sub-pillar	Flat	42	0.974	Step-wise search
Sub-pillar	Proportional	42	0.907	Exhaustive search
Sub-pillar	Proportional	42	0.977	Step-wise search

Table 16: Variable perspective: allocation rules compared, by level, budget and search method

Level	Allocation Rule	Indicators	r_m	r_m^2 (%)	Search Method
Pillar	Flat	21	0.854	72.9	Exhaustive search

Level	Allocation Rule	Indicators	r_m	r_m^2 (%)	Search Method
Pillar	Flat	21	0.885	78.3	Step-wise search
Pillar	Proportional	21	0.860	73.9	Exhaustive search
Pillar	Proportional	21	0.885	78.3	Step-wise search
Pillar	Flat	42	0.968	93.6	Step-wise search
Pillar	Proportional	42	0.963	92.6	Exhaustive search
Pillar	Proportional	42	0.971	94.3	Step-wise search
Sub-pillar	Flat	21	0.826	68.2	Exhaustive search
Sub-pillar	Flat	21	0.878	77.0	Step-wise search
Sub-pillar	Flat	42	0.966	93.2	Step-wise search
Sub-pillar	Proportional	42	0.946	89.5	Exhaustive search
Sub-pillar	Proportional	42	0.967	93.6	Step-wise search

Finding: Step-wise search scores at or above its exhaustive-search counterpart in every one of the five allocation/budget combinations both methods were run on, under both metrics (ten comparisons in total) - not a coincidence, but a consequence of what each method optimizes: exhaustive search maximizes a *local*, group-level criterion and is merely re-scored against the full dataset afterwards, while step-wise search optimizes the full-dataset criterion directly at every step (Methodology; discussed per-perspective above). Exhaustive search remains the only one of the two with a guarantee of finding the best possible subset for the criterion it is given; step-wise search offers no such guarantee, only a local optimum from a random start, mitigated here by keeping the best of several restarts.

Finding: Proportional allocation at 42 indicators outperforms flat allocation at 21 indicators on both metrics. Compared at a common budget, however, the two rules are nearly indistinguishable: at 21 indicators across pillars, proportional allocation reaches a rank correlation of 0.887 against flat allocation’s 0.886, and $r_m = 0.86$ against 0.854.

Budget therefore accounts for almost all of the apparent advantage. Holding the rule fixed and doubling the budget from 21 to 42 indicators raises the rank correlation by 0.068 and r_m by 0.103; holding the budget fixed and switching from flat to proportional changes them by 0.001 and 0.006. The budget effect is roughly two orders of magnitude larger on the distance metric and one on the PCA metric.

The case for proportional allocation is therefore representational rather than statistical: no group is systematically over- or under-weighted relative to its size in the index, at no measurable cost to either criterion. Whether indicator count is an appropriate basis for that weighting remains a judgement for the analyst, and the near-equivalence observed here need not hold for an index whose sub-pillars are more strongly correlated within pillars.

6 Conclusions

1. **Meaningful reduction is possible, with a clear methodological trade-off.** The headline 21-indicator subset (one reflective indicator per sub-pillar) captures r_m^2 68.2% of the full PCA’s variance structure and a Rand Index of 0.886 against the full clustering, at a 73% reduction in indicator count.
2. **The gain over random selection depends on the level at which indicators are allocated, not on the budget.** Allocating at pillar level exceeds the average random subset of matched size at both budgets (0.886 against 0.847 at 21 indicators; 0.955 against 0.932 at 42). Allocating at sub-pillar level falls below it at both. No constrained subset reaches the best random subset of its size, which is the cost of requiring every group to be represented.
3. **The allocation rule matters far less than the budget.** Compared at a common 21-indicator budget, flat and proportional allocation across pillars differ by 0.001 in rank correlation and 0.006 in r_m , while doubling the budget under a fixed rule gains 0.068 and 0.103 respectively. Proportional allocation’s benefit is representational rather than statistical.
4. **Recovery of the country clustering saturates well below the full indicator set.** The four candidate subsets (flat and proportional, by pillar and by sub-pillar) return Rand Indices within 0.029 of one another despite differing twofold in size and by 0.119 in rank correlation. Beyond covering every pillar and sub-pillar, further indicators refine the distance structure without changing which countries group together.
5. **The r_m criterion treats “information” as PCA variance, which may downweight indicators with low covariance but high substantive/policy relevance; the headline subset should be reviewed against domain expertise, not adopted on statistical grounds alone.**

7 Limitations

1. **Single-year analysis.** This analysis covers GII 2025 only; temporal stability of the recommended subset across years has not been assessed.
2. **PCA-defined “information.”** The r_m criterion equates information content with PCA variance structure; a substantively important but low covariance indicator could be excluded even if policy-relevant.
3. **Local search proxy.** Exhaustive search within each pillar/sub-pillar uses a local r_m computed on that group’s own sub-PCA (not the full 78-indicator PCA) as a tractable stand-in for an otherwise infeasible full search; the assembled subset is re-scored against the true full-dataset metric for reporting, but the *search* itself never directly optimizes that metric. Step-wise local search (Methodology) avoids this specific limitation, since it optimizes the full-dataset metric directly - at the cost of no longer guaranteeing the result

is the best subset achievable for that allocation, only a local optimum from a random start.

4. **Linear relationships assumed.** Both the rank-correlation and r_m criteria are built on linear/monotonic structure (PCA, Euclidean distance); non-linear relationships between indicators are not captured.
5. **Domain-specific validity not tested.** Whether the recommended subset remains meaningful to GII’s intended policy audience has not been evaluated here and would require domain-expert review.

8 Data Availability

All data and code are available in this repository:

- Raw data: `gii-data/idata.xlsx`, `gii-data/imeta.xlsx`
- Processed data: `data/gii_data_standardized.csv`, `data/gii_data_complete_cases.csv`
- Analysis scripts: `scripts/00_setup.R` through `scripts/04_variable_perspective.R`, plus `scripts/04_step-wise.R` (step-wise local search), shared functions in `scripts/R/`
- Results: `outputs/` (CSVs for tabular results, `.rds` for clustering objects; `stepwise_search_results.csv` and `stepwise_selected_variables.csv` for the step-wise search)

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10 Appendix

11 Data Imputation Documentation

Missing data is an inherent challenge in composite index construction. This analysis used k-Nearest Neighbors (kNN) imputation to address missing values before conducting the main analyses.

11.1 Imputation Process

- **Method:** k-Nearest Neighbors (kNN) imputation
- **Parameters:** $k = 5$ neighbors (VIM::kNN, other parameters left at package defaults)
- **Implementation:** R VIM package
- **Indicators Used:** Level 1 indicators from IMETA metadata

11.2 Missing Data Statistics

Table 17: Imputation Process Summary

Metric	Count
Total Missing Values	1211
Variables with Missing Data	58
Countries with Missing Data	135

Note: 1211 missing values across 58 of 78 indicators, affecting 135 of 139 countries, were imputed prior to standardization.

12 Descriptive Statistics Tables

12.1 Descriptive Statistics - Original Data

The table below presents descriptive statistics for all Level 1 indicators in the original dataset before imputation:

Table 18: Descriptive Statistics - Original Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Government effectiveness	0.15	0.03	-1.75	2.32	-0.71	0.29
Operational stability for businesses	2.23	2.15	0.70	4.45	0.03	0.58
Regulatory quality	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law	0.07	-0.12	-2.15	1.97	-0.77	0.29
Entrepreneurship policies and culture	4.72	4.72	2.92	7.25	-0.21	0.45
Policy stability for doing business	4.03	3.98	1.83	6.45	-0.69	0.02
Expenditure on education	4.25	4.16	0.35	9.39	0.80	0.29
Government funding/pupil, secondary	19.60	19.41	6.00	51.06	1.57	0.74
PISA scales in reading, maths and science	439.44	439.29	337.45	579.03	-0.84	0.07
Pupil-teacher ratio, secondary	15.89	13.53	6.14	68.13	9.84	2.46
School life expectancy	14.17	14.36	6.34	20.85	-0.23	-0.28
Graduates in science and engineering	23.29	23.14	9.56	41.10	-0.33	0.17
Tertiary enrolment	51.51	54.40	2.71	166.67	-0.02	0.37
Tertiary inbound mobility	7.34	4.24	0.00	69.73	15.40	3.35
QS university ranking	21.29	10.43	0.00	97.63	0.29	1.14
Global corporate R&D investors	926.35	0.00	0.00	36142.79	60.93	7.11
Gross expenditure on R&D	1.01	0.61	0.01	6.35	4.50	1.92
Researchers	2269.70	922.47	4.50	10413.00	0.53	1.21
Government's online service	0.68	0.72	0.17	1.00	-0.95	-0.40
ICT access	83.02	88.25	24.33	100.00	0.54	-1.12
ICT use	79.65	84.47	29.84	99.36	1.78	-1.49
Electricity output	4936.33	3183.45	29.76	53263.16	24.28	4.06
Gross capital formation	24.49	23.70	11.60	44.56	0.63	0.74
Logistics performance	3.12	3.10	2.10	4.30	-1.10	0.19
GDP/unit of energy use	12.23	11.06	2.16	41.21	4.01	1.59

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
Domestic credit to private sector	60.33	48.53	8.14	248.76	2.51	1.51
Finance for startups and scale-ups	4.49	4.49	1.99	6.90	-0.48	0.09
Loans from microfinance institutions	2.09	1.05	0.00	17.34	12.83	3.33
Joint venture deals	0.21	0.06	0.00	4.55	56.21	6.66
Late VC deals intensity	0.32	0.03	0.00	11.43	55.80	7.13
Market capitalization	80.85	37.45	0.20	1507.77	55.74	6.96
Venture capital investors	0.65	0.14	0.01	29.23	101.26	9.79
Venture capital recipients	0.24	0.08	0.01	5.50	67.12	7.42
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Domestic industry diversification	0.18	0.14	0.07	0.55	3.07	1.61
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Females employed w/advanced degrees	13.49	12.78	0.14	31.04	-1.14	0.26
GERD financed by business	34.31	38.44	0.00	80.84	-1.07	-0.07
GERD performed by business	0.77	0.38	0.00	5.90	7.67	2.33
Knowledge-intensive employment	27.07	23.94	2.02	68.01	-0.95	0.31
Patent families	1.30	0.03	0.00	73.67	114.09	10.30
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
State of cluster development	4.00	3.90	2.59	5.40	-0.79	0.24
WU Rankings industry and global outlook	55.74	49.40	26.70	97.10	-0.99	0.58
University–industry R&D collaboration	3.97	3.86	2.30	6.72	-0.12	0.38
FDI net inflows	3.93	2.50	- 161.34	224.25	66.45	3.27
High-tech imports	9.33	8.01	1.30	55.19	21.43	3.65
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
Research talent	33.25	32.27	0.30	81.91	-1.19	0.21
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Patents by origin	2.54	0.62	0.00	61.56	39.40	5.92
PCT patents by inventor origin	0.62	0.07	0.00	7.57	11.91	3.30
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Utility models by origin	1.55	0.16	0.00	88.51	84.62	9.17
Software spending	0.24	0.21	0.00	1.26	3.08	1.42
Labor productivity growth	0.83	0.86	-4.57	6.26	0.84	-0.01
High-tech manufacturing	25.19	21.99	1.04	79.65	0.15	0.75
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Production and export complexity	0.09	-0.05	-2.18	2.29	-0.52	0.23
High-tech exports	4.57	1.51	0.02	48.39	12.41	3.08
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Global brand value, top 5,000	3.59	0.78	0.00	21.12	1.78	1.64
Intangible asset intensity, top 15	41.66	56.72	-	89.79	28.96	-4.81
			371.30			
Industrial designs by origin	2.21	0.92	0.00	23.28	11.87	3.25
Trademarks by origin	36.72	28.95	0.00	202.33	7.36	2.24
Creative goods exports	1.27	0.38	0.00	11.50	7.00	2.60
Cultural and creative services exports	0.96	0.46	0.00	18.31	53.87	6.48
Entertainment and media market	0.93	0.47	0.02	3.59	-0.06	0.92
National feature films	5.01	2.67	0.00	46.93	18.95	3.70
Mobile app creation	1379627.73	181351.44	0.04	98744342.48	123.62	10.98
GitHub commits	1326116.27	410927.87	2571.55	14402260.97	14.47	2.99
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93

12.2 Descriptive Statistics - Complete Data

The table below presents descriptive statistics for all Level 1 indicators in the complete dataset after kNN imputation and case-wise cleaning:

Table 19: Descriptive Statistics - Complete Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Operational stability for businesses	2.23	2.15	0.70	4.45	0.03	0.58
Government effectiveness	0.15	0.03	-1.75	2.32	-0.71	0.29
Regulatory quality	0.16	0.11	-2.03	2.31	-0.72	0.21

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Rule of law	0.07	-0.12	-2.15	1.97	-0.77	0.29
Policy stability for doing business	4.03	3.98	1.83	6.45	-0.65	0.02
Entrepreneurship policies and culture	4.63	4.54	2.92	7.25	0.28	0.64
Expenditure on education	4.24	4.14	0.35	9.39	0.83	0.31
Government funding/pupil, secondary	18.48	17.96	6.00	51.06	1.96	0.85
School life expectancy	13.92	13.94	6.34	20.85	-0.42	-0.13
PISA scales in reading, maths and science	419.11	403.60	337.45	579.03	-0.43	0.70
Pupil-teacher ratio, secondary	15.80	13.28	6.14	68.13	10.01	2.48
Tertiary enrolment	50.58	53.42	2.71	166.67	0.00	0.41
Graduates in science and engineering	22.88	22.88	9.56	41.10	-0.31	0.25
Tertiary inbound mobility	6.61	3.14	0.00	69.73	17.76	3.58
Researchers	1922.18	669.77	4.50	10413.00	1.08	1.44
Gross expenditure on R&D	0.88	0.42	0.01	6.35	5.71	2.16
Global corporate R&D investors	933.55	0.00	0.00	36142.79	61.25	7.12
QS university ranking	21.29	10.43	0.00	97.63	0.29	1.14
ICT access	82.88	88.19	24.33	100.00	0.47	-1.09
ICT use	78.81	83.81	29.84	99.36	1.28	-1.36
Government's online service	0.68	0.73	0.17	1.00	-0.94	-0.41
Electricity output	4663.61	2879.19	29.76	53263.16	25.59	4.17
Logistics performance	3.04	2.90	2.10	4.30	-0.96	0.45
Gross capital formation	24.48	23.70	11.60	44.56	0.67	0.75
GDP/unit of energy use	12.00	10.93	2.16	41.21	4.46	1.69
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Finance for startups and scale-ups	4.25	4.11	1.99	6.90	-0.45	0.48
Domestic credit to private sector	60.08	48.30	8.14	248.76	2.47	1.51
Loans from microfinance institutions	1.43	0.63	0.00	17.34	26.25	4.59
Market capitalization	58.57	25.34	0.20	1507.77	88.80	8.73
Venture capital investors	0.56	0.11	0.01	29.23	120.91	10.68
Venture capital recipients	0.22	0.07	0.01	5.50	73.92	7.78
Late VC deals intensity	0.25	0.01	0.00	11.43	71.99	8.09
Joint venture deals	0.18	0.04	0.00	4.55	66.13	7.21

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic industry diversification	0.19	0.17	0.07	0.55	2.05	1.19
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Knowledge-intensive employment	26.67	23.31	2.02	68.01	-0.89	0.38
Females employed w/advanced degrees	13.06	11.20	0.14	31.04	-1.04	0.35
GERD performed by business	0.50	0.06	0.00	5.90	11.94	2.97
GERD financed by business	28.45	28.73	0.00	80.84	-1.02	0.34
Public research-industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
University-industry R&D collaboration	3.95	3.83	2.30	6.72	-0.05	0.42
WU Rankings industry and global outlook	51.65	42.20	26.70	97.10	-0.25	1.02
State of cluster development	3.99	3.84	2.59	5.40	-0.75	0.28
Patent families	1.29	0.03	0.00	73.67	114.91	10.33
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
High-tech imports	9.32	8.03	1.30	55.19	21.61	3.67
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
FDI net inflows	3.91	2.49	-	224.25	66.93	3.29
			161.34			
Research talent	23.49	11.34	0.30	81.91	-0.61	0.86
Patents by origin	2.51	0.61	0.00	61.56	39.97	5.96
PCT patents by inventor origin	0.58	0.06	0.00	7.57	13.22	3.45
Utility models by origin	1.05	0.16	0.00	88.51	136.59	11.64
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Labor productivity growth	0.83	0.86	-4.57	6.26	0.87	0.00
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Software spending	0.24	0.20	0.00	1.26	3.09	1.42
High-tech manufacturing	22.20	18.35	1.04	79.65	0.37	0.94
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
Production and export complexity	0.08	-0.06	-2.18	2.29	-0.43	0.26
High-tech exports	4.53	1.51	0.02	48.39	12.49	3.09
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Intangible asset intensity, top 15	32.37	39.35	-	89.79	34.44	-4.68
			371.30			

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Trademarks by origin	36.74	28.99	0.00	202.33	7.57	2.26
Global brand value, top 5,000	3.20	0.47	0.00	21.12	2.47	1.81
Industrial designs by origin	2.12	0.84	0.00	23.28	12.62	3.34
Cultural and creative services exports	0.86	0.42	0.00	18.31	60.72	6.87
National feature films	4.47	2.69	0.00	46.93	26.15	4.26
Entertainment and media market	0.54	0.17	0.02	3.59	2.74	1.84
Creative goods exports	1.27	0.38	0.00	11.50	7.07	2.61
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93
GitHub commits	1357449.63	411718.75	2571.55	14402260.97	13.92	2.91
Mobile app creation	1305696.24	155386.88	0.04	98744342.48	131.12	11.31